

Retrieval-Augmented Generation (RAG)

Sample knowledge-base document for a Retrieval-Augmented Generation project.

What is RAG?

Retrieval-Augmented Generation is a pattern that combines information retrieval with a language model. Instead of asking the model to answer only from its learned parameters, the system first retrieves relevant external text and gives that text to the model as context.

Why Use RAG?

RAG can improve factual grounding, allow answers to use private or domain-specific documents, and make it easier to update knowledge without retraining the language model. It can also provide source references when metadata is preserved.

Document Loading

The first stage is ingestion. Files such as PDFs are opened and converted into text documents. Useful metadata includes the original filename, page number, section, document type, and ingestion date.

Chunking

Long documents are split into smaller chunks because embedding models and language models have context limits. Chunk size and overlap affect retrieval quality. Too-large chunks may contain unrelated ideas, while too-small chunks may lose context.

Embeddings

An embedding model converts text into a numeric vector that represents semantic meaning. Texts with similar meaning tend to have vectors that are closer in the embedding space.

Vector Database

A vector database stores embeddings together with text and metadata. During retrieval, the question is embedded and compared with stored vectors. The closest chunks are returned as candidate context.

Generation

The retrieved chunks are inserted into a prompt along with the user question. The language model then generates an answer based on the supplied context. A strong prompt can instruct the model to admit when the answer is not present in the documents.

Typical RAG Pipeline

A basic pipeline is: load documents, clean text, split into chunks, create embeddings, store vectors, embed the user query, retrieve top matching chunks, build a prompt, call the language model, and return the answer with sources.

RAG Evaluation

RAG evaluation includes retrieval quality and answer quality. Retrieval can be measured by whether relevant chunks appear in the top results. Answers can be evaluated for faithfulness, correctness, relevance, citation quality, and completeness.